

Data leakage jeopardizes ecological applications of machine learning

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Machine learning is a popular tool in ecology but many scientific applications suffer from data leakage, causing misleading results. We highlight common pitfalls in ecological machine-learning methods and argue that discipline-specific model info sheets must be developed to aid in model evaluations.

Machine learning has gained tremendous interest in ecology as a powerful tool for uncovering patterns and making data-based predictions^{1,2}. However, it can fail to yield generalizable predictions and answers when misapplied. Here, we explain data leakage³, which is a widespread cause of incorrect results in scientific applications of supervised machine learning. Addressing both ecologists who use machine learning and readers of ecology studies that are based on machine learning, we ask how researchers can avoid data leakage and how readers can evaluate the limitations of such studies.

Data leakage

Data leakage occurs when a model accesses information about the response variable that is absent in the model's intended application⁴ (Fig. 1). For example, the doctoral thesis of one of us (A.S.) involved training models to predict coastal biodiversity on the basis of a mix of human uses and environmental variables. This study initially took machine learning methods straight from a textbook and evaluated them with ten-fold cross-validation, and seemingly achieved excellent prediction accuracy. But the data were clustered around some field sites (for example, in protected areas). Therefore, the models had already 'seen' data collected near the test observations that were very similar owing to spatial autocorrelation. Noticing this problem and evaluating the models more appropriately, their prediction accuracy was much lower⁵. As this example illustrates, ecological data collection and preprocessing practices can create spurious relationships between features and response that are limited to the specific dataset. Relying on such relationships, models can make accurate predictions for the test set but fail when applied to new data. A known pitfall among technical specialists, data leakage is widespread in scientific applications: a recent review found it to affect over 300 published studies from 17 academic disciplines and warned of a potential reproducibility crisis³.

Shortcut learning⁶ is a type of data leakage and a powerful lens for understanding why models might fail to learn generalizable solutions. Consider a model that classifies animals in photographs and suppose all photographs of sheep in the labelled dataset have a green landscape as a background. A model trained with such data can learn to recognize sheep as objects against a green background (a shortcut) instead of recognizing their shape as humans do⁷ (a generalizable solution).

This model would fail to recognize sheep in other landscapes (for example, on snow). Yet, this model would appear to be capable of accurately identifying sheep if the training and testing data were sampled from the same biased collection of images.

Ecological examples

Below, we summarize three example studies that illustrate how data leakage can play out in ecological research.

The first study classified animals in camera trap images⁸. It shows that computer vision algorithms can make less accurate predictions when applied in new locations. For this purpose, the study compared how accurately models trained in some locations classified animals in new images from the same locations versus others. The species were always the same, but the context of the images (for example, background and lighting) differed. Errors for images from non-training locations were much larger than for held-out images from locations used in training. The model appeared to base predictions at least partly on local relationships between the animals and the image background (that is, on shortcuts) instead of generalizable properties of the animals, such as shape⁶.

The second study predicted kelp habitat on the basis of biophysical variables such as water depth and salinity⁹. Testing increasingly complex models, it shows how data leakage affects traditional supervised-learning approaches for species distribution modelling. Data on kelp presence came from a field survey that was specific to this study. With three-fold cross-validation, more complex models appeared to make more accurate predictions. But the simpler models were more accurate when tested against data from independent sources (a government database and satellite imagery). This difference in prediction skill on randomly sampled testing data versus data from other sources occurred because the training and testing data shared the same context (for example, place, time or population) and thus the same biases and potential shortcuts.

The third study trained simple linear models and basic machine-learning models such as random forests to predict chlorophyll *a* concentration from satellite measurements of ocean colour¹⁰. It shows that spatially biased and autocorrelated data cause misleading results. Instead of using a single labelled dataset, the study created 2,000 artificial datasets with chlorophyll concentrations from biogeochemical simulations instead of field measurements. The 'true' concentrations were thus known everywhere. When labelled data were intentionally limited to locations for which real field measurements existed, machine learning models had larger errors than a simple linear model (Supplementary Fig. 1a). However, ten-fold cross-validation incorrectly suggested the opposite. It underestimated the errors of the machine learning methods, making these models appear better (Supplementary Fig. 1b). Spatial block cross-validation¹¹ (in which whole subregions of the study area served as folds) reduced, but could not eliminate, these biases (Supplementary Fig. 1c). By contrast, when repeating the analyses with labelled data sampled from random locations, prediction errors

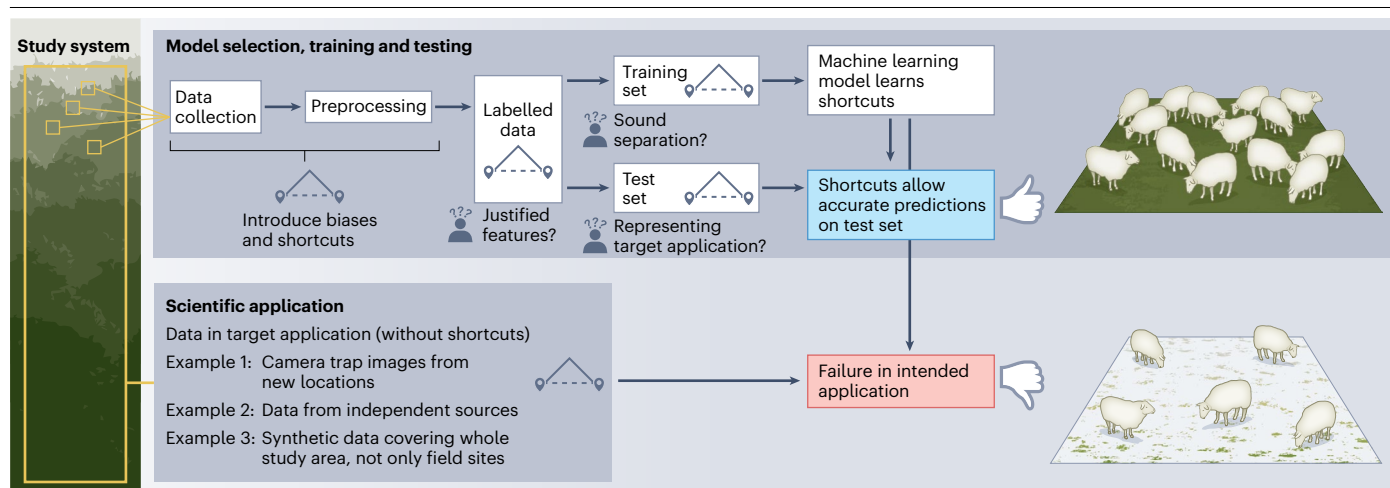


Fig. 1 | How data leakage might occur in ecological applications, explained through the lens of shortcut learning. Many machine learning applications in ecology involve supervised learning: models learn from so-called labelled data, for which both the desired output (response) and inputs (features or predictors) are known. For example, computer vision algorithms predict what an image shows (for example, which species) on the basis of features constructed from the image's pixels¹⁹. The labelled data are often split into a training set for fitting the model and a test set for estimating its prediction accuracy on previously unseen

examples (upper part of the figure). Many studies do this split repeatedly in cross-validation. For instance, k -fold cross-validation randomly breaks the data into k chunks (folds). Each fold is the test set for a model trained with the other $(k - 1)$ folds, and the k resulting error estimates are averaged. Finally, to make predictions or answer questions about a broader ecosystem, the model is applied to new data for which only the predictors are known (lower part of the figure). Because of data leakage, models can make accurate predictions for the test set yet fail in their intended ecological application.

were much lower, machine learning did outperform simpler models and ten-fold cross-validation yielded accurate error estimates.

These examples illustrate four points. First, models based on supervised learning made accurate predictions when evaluated with data from the same context. However, mirroring concerns about shortcut learning⁶, distribution shift¹² and model transferability to new geographical locations or taxa¹³, tests with data from different contexts revealed lower prediction skill. Second, error estimates biased by data leakage favoured more complex over simpler models. Echoing issues in ecological statistics¹¹ and other disciplines³, this result was misleading: simpler models generalized better to data from new contexts. Third, it is crucial to align model evaluations with the intended scope of the scientific application. For example, a model trained for a study region that is well-described by the data does not need to work well for other regions unless used to make predictions or scientific claims beyond the study region. By contrast, if making such predictions or claims, evaluating the model with data from new regions is essential. Fourth, opportunities for shortcuts (and hence for hard-to-detect data leakage) exist in many ecological applications. For example, whenever supervised learning is applied to spatially autocorrelated data (for which nearby observations are similar), models can learn to recognize geographical locations and predict values typical for them instead of learning the intended relationships between variables. Therefore, data that are biased spatially or temporally are a critical driver of data leakage in broad-scale ecological studies.

Addressing data leakage in ecology

Data leakage in ecological applications can be addressed at two levels. First, researchers who use machine learning can avoid data leakage with sound research designs and methods. Below, we provide fundamental methodological guidance. Second, journals can require standardized documentation of how a study avoided data leakage. Adopting these

solutions would raise the bar for the quality of published studies based on machine learning and enable readers who are not technical specialists to better understand the limitations of a study. The next paragraphs outline how these solutions can build on existing tools.

Researchers can use various approaches to address data leakage (Table 1). For example, if one is aware of this pitfall, it is straightforward to exclude the test set in early data processing and modelling steps. Other types of data leakage are more challenging to address and require domain knowledge – especially in ecology, for which issues such as data independence are tied to the scale of analysis¹⁴. For example, models are ideally evaluated with independent data that represent the intended application³. If such data are unavailable, block cross-validation is a good alternative because ecological data often have dependence structures (for example, spatial autocorrelation)¹¹. An R package for block cross-validation is available¹⁵. Differences between the error estimates of different cross-validation strategies can inform a discussion of the model's proper scope of application (for example, differences in prediction errors for locations that are well-covered by training data versus sites with no data nearby).

Unfortunately, such solutions for widespread data leakage problems remain underused. For example, many articles lack a detailed description of methods, such as the exact sequence of analysis steps. Besides the resulting lack of computational reproducibility¹⁶, such missing details make it impossible to rule out even simple types of data leakage. Because researchers have incentives to publish promising results without further questioning their data and methods^{17,18}, it is essential to empower editors, reviewers and readers to better evaluate manuscripts that are based on machine learning. We hence support requiring model info sheets with studies that are based on supervised learning³.

Model info sheets are checklists in which authors answer questions and provide justifications to rule out statistical pitfalls. Filling out and evaluating model info sheets involves additional work during

Table 1 | Data leakage in ecology

Type	Explanation and ecological examples	Solutions
Insufficient separation of training and test set	The test set is used in preprocessing and modelling steps such as filling data gaps and selecting features. This leads to underestimated errors because these modelling steps use information about correlations in the test set.	Exclude test set from all modelling steps. In cross-validation, avoid steps such as gap-filling and feature selection before cross-validation. Instead, repeat them within the cross-validation. Separate training and test sets sufficiently in space and time ¹¹ .
Use of illegitimate features	The study uses features that are influenced by the output variable. Examples include (1) using per cent of agricultural land to predict forest cover in an area, because the more agricultural land there is, the less space remains for other land cover types; (2) including geographical coordinates or day-of-year as features in a model that is then used to predict ecological effects of future climate change. Here, the training and test sets will include relationships describing seasonal and spatial patterns that are useful predictors now but likely to change in the intended application.	Select and justify all features with domain knowledge; justify the assumption of stationarity (that statistical relationships between variables are likely to remain the same) when extrapolating in space or time.
Test set is not representative of the scientific application	The test set does not represent the intended scientific application. Examples include (1) including training data from later times than testing data for a model to make predictions about the future; (2) training and testing models with data from the same locations (for example, a few field sites with clusters of data) but making predictions for other locations (for example, regional-scale mapping).	Use test sets and error estimation methods tailored to the scientific problem and target scale of analysis. Suitable strategies may include block cross-validation ¹¹ , independent data validation ⁹ and other out-of-distribution tests ⁶ .

This table outlines ecological examples of, and solutions to, common data leakage types that have been identified in other scientific disciplines³.

manuscript preparation and review. Yet, as reviewers, we often find that ruling out data leakage requires additional information from authors, adding an unnecessary round of revisions and sometimes – if data leakage was present – changes to a manuscript's conclusions. Thus, using model info sheets to guide modelling, writing and peer review simplifies the preparation and publication of high-quality papers.

A generic template for model info sheets developed by computer scientists is available [online](#). In our view, this template asks all the right questions while also requiring a comprehensive understanding of data leakage in ecological applications. For example, one question asks how the training–testing data split was observed when evaluating the model. To answer this question, modellers, reviewers and readers must know what structures in ecological data jeopardize a clean split (for example, similar observations from close-by locations or similar

behaviour by animals in groups)¹¹ – causes of data leakage that many studies ignore. Therefore, we encourage the ecological research community to develop a discipline-specific template that serves as a guide to typical data leakage pitfalls in ecology. To incorporate the breadth of ecological machine-learning applications, such a template could be developed (for example) in workshops at professional society meetings. Until then, Table 1 might serve as a general guide that lists the main points to address.

Although supervised learning has a long history in ecology, the trend towards more complex machine-learning approaches increases the risk of misleading results due to data leakage. With more flexibility to exploit shortcuts and difficult interpretation, such models often outperform simpler models in standard tests yet perform no better or worse once data leakage is accounted for^{3,10}. With this trend in mind, changes in the evaluation and documentation of supervised learning methods in ecology are urgently needed. From individual researchers to journal policies, making these changes will require a concerted effort. Yet, if data leakage creates a reproducibility crisis in machine learning-based science³, we must address this challenge in ecology now.

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